

Lab Exercise #8: Update Stock Item and Price

1 Overview

In this lab, write a C program which takes an item's name, price and quantity stores them in a structure. The program then ask for a price change and quantity change and passes the information to a function. The program prints the item name, the new price, the total quantity and the total value of the total quantity of the item at the new price.

Your program should

- Request the user to input an item name, price and quantity. Read in each of these inputs.
- Store the name, price and quantity in a structure.
- Request the user to input a price and quantity change. Read in each of these inputs.
- Use a function that has all parts of the structure as parameters to update the price and quantity. The function should return the entire structure.
- Save the returned structure as a new variable.
- The program should output the part name, the new price and the new quantity.
- The program should print out the total value of the number of items on hand at the new price.

2 Code Documentation

The very first lines in *all* of your programming project files this semester should be comments in the class header file (copy from header.c). Of course, the items in the square brackets [] in the comments should be filled in with the proper information.

In addition, the first output for all of your projects this semester, both for the lab exercises and the programming assignments, should be the name of the project and your name, e.g:

```
printf("Lab #8 - Joe Student\n");
```

Obviously, for the lab exercises, the output should include the names of both students in the lab group, e.g: `printf("Lab #8 - Joe Student and Jane Doe\n");`

3 Sample Run

Your output should match the sample run below.

```
Lab #9: Mark Schwitzerlett and Leonard Euler
```

```
Input the item name: Printer
```

```
Input the item price: 349.99
```

```
Input the item quantity: 5
```

```
Input price increment value: -38.60
```

```
Input quantity increment value: 7
```

```
Updated values of item
```

```
Name : Printer
```

```
Price : 311.39
```

```
Quantity : 12
```

```
Value of the item = 3736.68
```

```
...Program finished with exit code 0
```

```
Press ENTER to exit console.□
```

4 Deliverables

You should turn in a stand-alone, complete application program (your C source code, a single “.c” file) containing a `main()` function. All labs and projects this semester will be submitted in Gradescope (via a Canvas assignment).

Your file should be named `Lab#8_FirstName_LastName.c`

Due date: The lab is due at the end of your lab period, either Monday, April 6th or Tuesday, April 7th.

Be sure to save a copy of your .c file.